

Commonly used contrast agents include gadolinium-based compounds. Depending on its environment, gadolinium (Gd) can have eight or nine sites in its coordination sphere, allowing Gd^{3+} to interact with eight water molecules as in $\text{Gd}(\text{H}_2\text{O})_8^{3+}$. The electronic configuration of gadolinium is shown in Figure 1.



Figure 2 Gd-DOTA with a water coordinated to Gd^{3+}

Figure 3 Gd-daa3 when Zn^{2+} is present

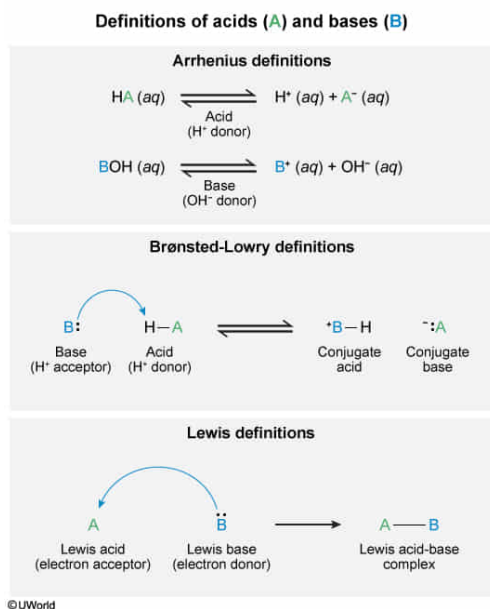
When Gd^{3+} binds to DOTA, Gd^{3+} is acting as a(n):

- ✔ ☒ A. Lewis acid. [66%]
☐ B. Lewis base. [22%]
☐ C. Arrhenius acid. [6%]
☐ D. Arrhenius base. [4%]

Correct

65% answered correctly

Explanation:



Acids and bases can be defined in different ways. According to the **Lewis definitions** of acids and bases, **bases** are **electron donors** whereas **acids** are **electron acceptors**. These definitions are **more general** than the **Arrhenius** and **Brønsted-Lowry** definitions, which involve the exchange of hydrogen ions (H^+ , or protons) and/or hydroxide ions (OH^-). According to the Arrhenius definitions, an acid is an H^+ donor and a base is an OH^- donor; according to the Brønsted-Lowry definitions, an acid is an H^+ donor and a base is an H^+ acceptor.

In **coordination chemistry**, **metal ions** such as Gd^{3+} **act as Lewis acids** because they have empty **valence orbitals** that can readily accept electrons. Molecules or atoms with lone pairs of electrons act as

Lewis bases and *coordinate* to the metal ion, forming a complex ion; these electron-donating groups are referred to as ligands. In this example, the ligand molecules are amide nitrogen residues and water molecules.

(Choice B) A Lewis base is an electron donor. In coordination chemistry, the Lewis bases are the ligands that coordinate with the metal ion.

(Choices C and D) The Arrhenius definitions of acids and bases involve the exchange of OH^- and H^+ ions. Because no hydrogen or hydroxide ions are involved in complexes formed between Gd^{3+} and DOTA, neither species acts as an Arrhenius acid or base.

Educational objective:

Lewis bases are electron donors whereas Lewis acids are electron acceptors. Lewis definitions differ from the Arrhenius definitions, in which an acid is an H^+ donor and a base is an OH^- donor, and from Brønsted-Lowry definitions, in which an acid is an H^+ donor and a base is an H^+ acceptor.

Time spent:0 sec
Subject:
General Chemistry

QID:400220
System:
Atomic Theory and Chemical Composition

Magnetic resonance imaging (MRI) interprets the *nuclear relaxation* or *relaxivity* of hydrogen nuclei in resident water molecules. Hydrogen nuclei in water have two magnetic spin states: alpha and beta. In the presence of an applied magnetic field, these spins align with the field in an "excited" state and then relax back to the "ground" state, producing a signal with intensity proportional to their relaxivity. Frequently, an MRI contrast agent is used to increase the relaxation time in coordinated molecules, resulting in a more intense signal.

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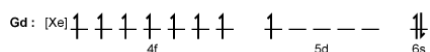


Figure 1 Electronic configuration of gadolinium

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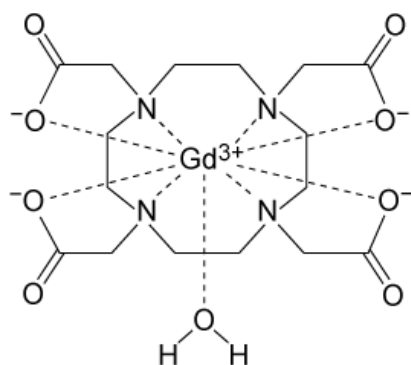


Figure 2 Gd-DOTA with a water coordinated to Gd^{3+}

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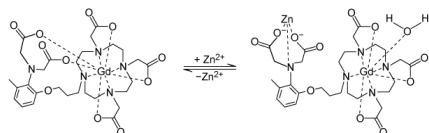


Figure 3 Gd-daa3 when Zn^{2+} is present

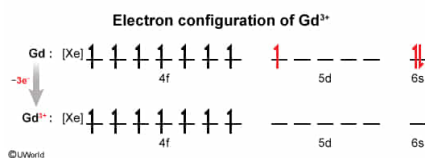
Gadolinium becomes ionized to Gd^{3+} when it loses electrons from which orbital(s)?

- ☐ A. 4f, 4d, and 5s [1%]
- ☐ B. 4f [25%]
- ☐ C. 4f, 5d, and 6s [7%]
- ☒ D. 5d and 6s [65%]

Correct

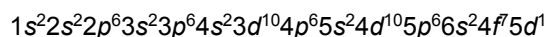
64% answered correctly

Explanation:

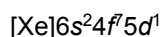


The **electron configuration** of an atom describes the **placement of electrons** within the atom and can be used to determine the order of electron removal during ionization. In standard notation, the electron shell is denoted by the principal quantum number (n), and subshells are typically denoted by a letter (eg, s, p, d, f). The principal quantum number is related to the distance of an orbital from the nucleus whereas the subshells are related to **orbital shape**.

The full electron configuration of an atom lists all orbitals and the number of electrons occupying each. However, these lists can become cumbersome for elements with many electrons, so a shorthand version is often used in which the preceding **noble gas** on the **periodic table** is listed followed by the remainder of the configuration. For instance, the electron configuration for gadolinium can be reduced from the long form:



to the much simpler noble gas form:



Note here that Gd shows an **exception** to the usual **orbital filling order**. Because the energy difference between the 4f and 5d orbitals is relatively small, an electron in the 4f orbital is promoted to the 5d orbital to minimize electronic repulsion. This in turn stabilizes the electron.

During **ionization**, electrons are **removed** from the **valence electron shell** with the greatest principal quantum number first because these electrons are farthest from the nucleus and are therefore the least tightly bound. Once all electrons from that shell are removed, additional electrons are removed from the shell with the next greatest principal quantum number. Therefore, gadolinium will lose its first two electrons from the **6s** subshell followed by **one electron** from the **5d** subshell to produce Gd³⁺, which has the configuration [Xe]4f⁷.

(Choices A, B, and C) Gadolinium does not lose any electrons from the 4f orbital. Atoms tend to lose the outermost valence electrons first because these electrons are the least tightly bound.

Educational objective:

The electron configuration of an atom describes the placement of all the electrons within the atom. During ionization, electrons are removed first from the orbital with the greatest principal quantum number.

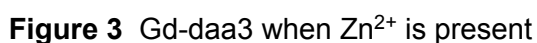
Time spent: 0 sec
Subject:
 General Chemistry

QID: 400221
System:
 Atomic Theory and Chemical Composition

Commonly used contrast agents include gadolinium-based compounds. Depending on its environment, gadolinium (Gd) can have eight or nine sites in its coordination sphere, allowing Gd^{3+} to interact with eight water molecules as in $\text{Gd}(\text{H}_2\text{O})_8^{3+}$. The electronic configuration of gadolinium is shown in Figure 1.



Zinc ions (Zn^{2+}) are required to store insulin and are co-released when insulin is secreted by the pancreas. To detect zinc ions in the body, researchers have designed a variation of Gd-DOTA called Gd-daa3 with two diaminoacetate (daa) arms that preferentially bind to Zn^{2+} ions (Figure 3). When Zn^{2+} is absent, these arms coordinate to Gd^{3+} and create a nine-coordinate complex that prohibits water from binding to Gd^{3+} . Binding of diaminoacetate arms to Zn^{2+} frees Gd^{3+} to coordinate with water molecules. This configuration results in increased signal intensity in the parts of the cell where Zn^{2+} is located.



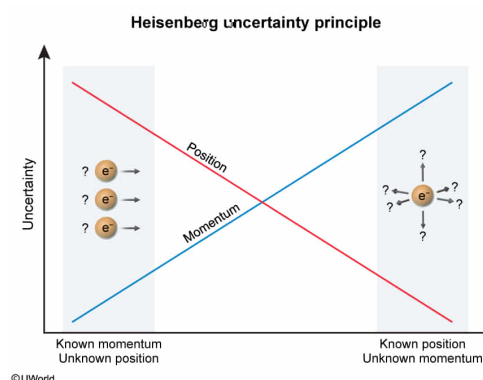
The Heisenberg uncertainty principle impacts the study of nuclei and small particles such as electrons. Which of the following is a consequence of this principle?

- ☐ A. Two electrons in the same orbital cannot both have parallel alignment with the magnetic field. [18%]
- ☐ B. Neutrons and protons are affected by the nuclear force almost identically. [1%]
- ☒ C. Electrons in coordination bonds can be described only as probability distributions. [74%]
- ☐ D. Particles release energy when transitioning from an excited state to a ground state. [6%]

Correct

73% answered correctly

Explanation:



The **Heisenberg uncertainty principle** states that the exact position and momentum of a particle cannot both be known at the same time. In fact, the **uncertainty of position and momentum** is **inversely proportional**: the more accurately an object's position is known, the less accurately its momentum can be known, and vice versa.

This principle most strongly impacts the study of small particles such as electrons. Its acceptance ended scientists' attempts to describe the exact path of an electron in its orbital, as understanding this path would require simultaneous knowledge of the electron's position and momentum. Instead, electron orbitals are now modeled as **probability distributions** in which electrons are most likely to be found.

(Choice A) Although it is true that two electrons in the same orbital must have spins of opposite alignment, this fact is related to the **Pauli exclusion principle**, not the Heisenberg uncertainty principle. The Pauli exclusion principle says that no two electrons in a given atom can have the same set of quantum numbers.

(Choice B) The nuclear force is the attractive force that binds neutrons and protons in the nucleus. Because neutrons and protons have approximately the same mass, they are influenced similarly by the nuclear force; however, this is not related to the Heisenberg uncertainty principle.

(Choice D) Energy is released when particles transition from a higher energy state (ie, excited state) to a lower energy state (ie, ground state). However, this phenomenon is due to the conservation of energy rather than the Heisenberg uncertainty principle.

Educational objective:

The Heisenberg uncertainty principle states that the position and momentum of a particle cannot both be precisely known at the same time. The uncertainty of the position and the uncertainty of the momentum are inversely proportional.

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General Chemistry

QID: 400222

System:
Atomic Theory and Chemical Composition

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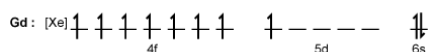


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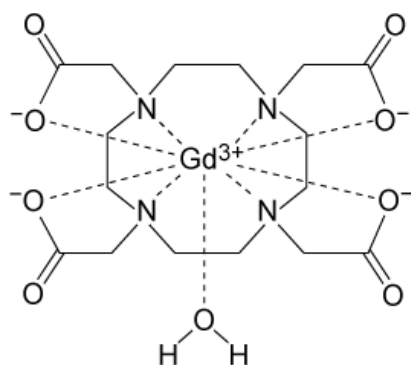


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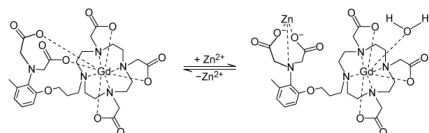


Figure 3 Gd-daa3 when Zn^{2+} is present

Which of the following accurately describes the magnetic properties of Gd^{3+} ions?

- ☐ A. Gd^{3+} is diamagnetic because of unpaired electrons. [14%]
- ☒ B. Gd^{3+} is paramagnetic, and its electron spins align parallel to the applied magnetic field. [67%]
- ☐ C. Gd^{3+} is diamagnetic because ions are attracted to the magnetic field due to their positive charge. [6%]
- ☐ D. Gd^{3+} is paramagnetic, and therefore repels magnetic field lines. [11%]

Correct

66% answered correctly

Explanation:

Types of magnetism		
	Diamagnetic	Paramagnetic
Electron pairing	 No unpaired electrons	 At least one unpaired electron
Spin alignment with magnetic field B	 Anti-parallel	 Parallel
Reaction to magnets	 Very weakly repelled	 Attracted
Effect on magnetic field lines	 Field bends slightly away from the material	 Field bends toward the material

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Hund's rule states that the order of orbital filling should **maximize** the number of **unpaired electrons**. Ferromagnetic and **paramagnetic** atoms and molecules have **one or more unpaired electrons** that can **align with a magnetic field**. Therefore, these atoms are attracted to magnets, and paramagnetic materials make good MRI contrast agents by enhancing the signal in tissues in which they have accumulated.

Alternatively, in **diamagnetic** atoms and molecules all **electrons are spin-paired** (ie, all orbitals contain two electrons). Diamagnetic materials **weakly repel** magnets, and their magnetic moments **align antiparallel** to an applied magnetic field. Noble gases such as **neon** are diamagnetic, as are **ions** that have lost their outer shell unpaired electrons.

As shown in the passage, gadolinium has seven unpaired electrons in the *4f* subshell, making Gd^{3+} **strongly paramagnetic**. Therefore, the unpaired electrons in Gd^{3+} align with an applied magnetic field.

(Choice A) Diamagnetic atoms and molecules do not contain unpaired electrons. Gd^{3+} is paramagnetic.

(Choice C) Gd^{3+} is paramagnetic (not diamagnetic), and the net charge of an atom or molecule does not influence its interaction with a magnetic field.

(Choice D) Gd^{3+} is paramagnetic, which means that it aligns with and is attracted to magnetic fields.

Educational objective:

Hund's rule states that orbital filling maximizes the number of unpaired electrons. Paramagnetic atoms and molecules have unpaired electrons that align parallel to an applied magnetic field; paramagnetic species are weakly attracted to magnets. In contrast, diamagnetic atoms and molecules have no unpaired electrons, and these species are repelled by magnets.

Time spent: 0 sec
Subject:
General Chemistry

QID: 400223
System:
Atomic Theory and Chemical Composition

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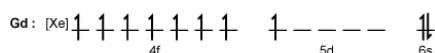


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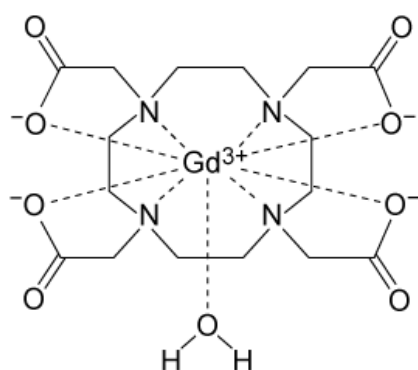


Figure 2 Gd-DOTA with a water coordinated to Gd^{3+}

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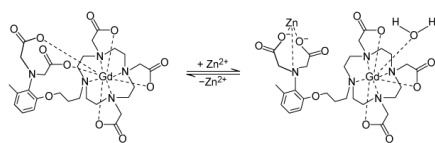


Figure 3 Gd-daa3 when Zn^{2+} is present

To determine whether Gd-daa3 is selective for Zn^{2+} , researchers compared relaxivity measurements of Gd-daa3 in the presence of different cations. Varying amounts of XCl_2 (where X is Zn^{2+} , Ca^{2+} , or Mg^{2+}) were added to a buffered 0.1 mM solution of Gd-daa3 at pH 7.4. What is the most appropriate control for this experiment?

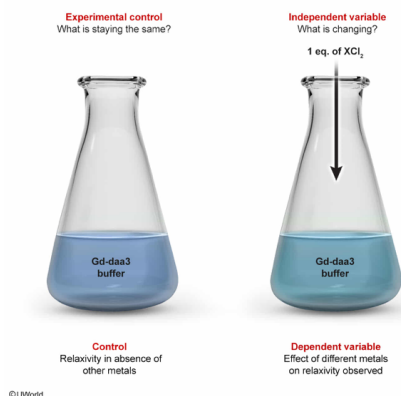
- ✓ ☒ A. Relaxivity measurements without XCl_2 [61%]
☐ B. Relaxivity measurements with ZnCl_2 [14%]

- ☐ C. Relaxivity measurements with BeCl_2 [5%]
- ☐ D. Relaxivity measurements without Gd-daa3 [18%]

Correct

60% answered correctly

Explanation:



An **experimental control** is a measurement taken with all **experimental conditions** in place **except** the **independent variables**. The dependent variables are unknown values measured and analyzed in relation to the independent variable. The control setup allows for **baseline measurements** of dependent variables in the **absence** of independent variables, allowing for a more accurate assessment of the independent variables' effects.

For these testing conditions, the independent variables include the amount of XCl_2 and the identity of X, and the dependent variable is relaxivity. The most appropriate experimental control includes **baseline readings of the relaxivity** of the Gd-daa3 solution in the **absence of XCl_2** . Therefore, the control solution contains only Gd-daa3 and buffer at pH 7.4. All instrumental settings should be the same for all experimental trials.

(Choice B) Because Zn^{2+} is one of the ions tested in this experiment (ie, an independent variable), ZnCl_2 would not be part of the control.

(Choice C) Even though BeCl_2 is not one of the ions tested in this experiment, it would not be included in the control. Be^{2+} ions are not present in all testing samples, and therefore could skew baseline relaxivity readings.

(Choice D) A control without Gd-daa3 would not provide baseline relaxivity measurements in the absence of the independent variable (ie, the amount of XCl_2 for each X).

Educational objective:

The experimental control is a measurement taken with all experimental conditions in place except the independent variable of interest. Controls provide baseline measurements so that the effects of independent variables on dependent variables can be accurately determined.

Time spent: 0 sec
Subject:
General Chemistry

QID: 400224
System:
Atomic Theory and Chemical Composition

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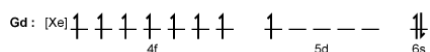


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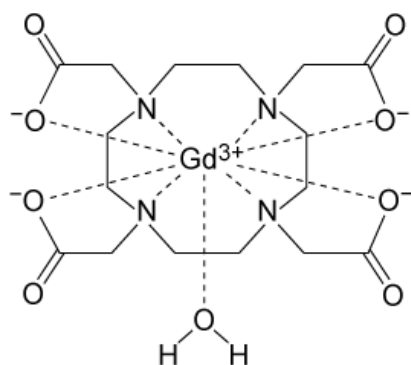


Figure 2 Gd-DOTA with a water coordinated to Gd^{3+}

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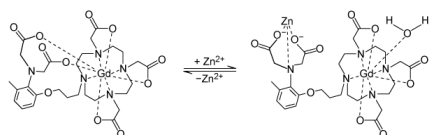


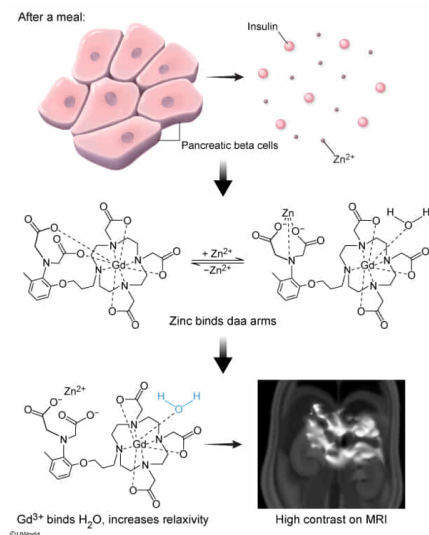
Figure 3 Gd-daa3 when Zn^{2+} is present

Two MRI images of the pancreas were taken at different times (Time A and Time B) in a mouse injected with Gd-daa3. Greater relaxivity was detected at Time B, suggesting all the following EXCEPT:

- ☐ A. Time B was soon after a meal when insulin is released. [13%]
- ☐ B. Fewer diaminoacetate arms were bound to Gd^{3+} ions at Time B than at Time A. [28%]
- ☐ C. The relaxivity of water molecule nuclei was increased at Time B. [18%]
- ☒ D. Gd^{3+} ions are bound to Zn^{2+} ions at Time B. [39%]

Correct
39% answered correctly

Explanation:



The passage describes the mechanism by which Gd-daa3 acts as a **contrast agent** in the **presence of zinc ions**. Zinc preferentially binds to diaminoacetate (daa), not Gd^{3+} , allowing Gd^{3+} to bind with water molecules and increase relaxivity. An increase in relaxivity at Time B suggests an increase in the release of zinc ions. Because Zn^{2+} is co-released with insulin, the image at Time B was likely taken shortly after a meal, when insulin is released from pancreatic beta cells (**Choice A**).

When Zn^{2+} binds to the diaminoacetate arms on Gd-daa3, gadolinium is released. Consequently, fewer diaminoacetate arms are bound to Gd^{3+} at Time B because more Zn^{2+} ions are present at Time B than at Time A (**Choice B**).

The released Gd^{3+} ions are then free to coordinate with water molecules, which increases the relaxivity of nuclei in water molecules (**Choice C**). This is the mechanism underlying the enhanced MRI signal.

Therefore, the greater relaxivity detected at Time B suggests the conclusions in all the answer choices EXCEPT (**Choice D**) because Gd^{3+} ions bind to water molecules, *not* Zn^{2+} ions.

Educational objective:

Contrast agents increase MRI signals by increasing the relaxivity of water molecules. These agents can be used to monitor biological processes if their mechanisms involve biochemical changes associated with those processes.

Time spent: 0 sec
Subject:
General Chemistry

QID: 400225
System:
Atomic Theory and Chemical Composition